HOMEWORK 6

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8.1. True or false:

a) Every change of coordinate matrix is square;
b) Every chance of coordinate matrix is invertible;
c) The matrices A and B are called similar if $B = Q^T AQ$ for some matrix Q;
d) The matrices A and B are called similar if $B = Q^{-1}AQ$ for some matrix Q;
e) Similar matrices do not need to be square.

Solution. TTFTF ■

8.2. Consider the system of vectors

$$(1, 2, 1, 1)^T, (0, 1, 3, 1)^T, (0, 3, 2, 0)^T, (0, 1, 0, 0)^T.$$

a) Prove that it is a basis in $\mathbb{F}^4$. Try to do minimal amount of computations.
b) Find the change of coordinate matrix that changes the coordinates in this basis to the standard coordinates in $\mathbb{F}^4$ (i.e to the coordinates in the standard basis $\mathbf{e}_1, \ldots, \mathbf{e}_4$).

Proof. Let v_1, v_2, v_3, v_4 be the vectors shown above respectively. Since they form a 4×4 matrix, it is sufficient to show that $av_1 + bv_2 + cv_3 + dv_4 = 0$ implies linear independence. We see $a = 0$ for the first row; $a + b = 0 \implies b = 0$ for the 4th row; $a + 3b + 2c = 0 \implies c = 0$ for row 3; $2a + b + 3c + d = 0 \implies c = 0$ for 2nd row. Thus, these vectors are a basis for $\mathbb{F}^4$.

$$[I]_{\mathcal{SB}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 1 \\ 1 & 3 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

■

8.3. Find the change of coordinates matrix that changes the coordinates in the basis $1, 1+t$ in $\mathbb{P}_1$ to the coordinates in the basis $1 - t, 2t$.

Solution. Let $\mathcal{A} = \{1, 1 + t\}$ and $\mathcal{B} = \{1 - t, 2t\}$. $1 = a(1 - t) + b(2t) \implies a = 1, b = 1/2$. $1 + t = a(1 - t) + b(2t) \implies a = 1, b = 1$. Hence,

$$[I]_{\mathcal{BA}} = \begin{pmatrix} 1 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}$$

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Date: October 13, 2026.

8.4. Let T be the linear operator in F^2 defined by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x - 2y \end{pmatrix}$$

Find the matrix of T in the standard basis and in the basis

$$(1,1)^T, (1,2)^T$$

Solution. Let $[T]_{\mathcal{S}} = \begin{pmatrix} 3 & 1 \\ 1 & -2 \end{pmatrix}$. Let $\mathcal{B} = \{(1,1)^T, (1,2)^T\}$. Let $\mathcal{S} = [I]_{\mathcal{SB}} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \implies \mathcal{S}^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. Thus $[T]_{\mathcal{B}} = \mathcal{S}^{-1}[T]_{\mathcal{S}}\mathcal{S} = \begin{pmatrix} 9 & 13 \\ -5 & -8 \end{pmatrix}$ ■

8.5. Prove, that if A and B are similar matrices, then $\operatorname{tr} A = \operatorname{tr} B$.

Proof. $B = Q^{-1}AQ$ by similarity. $\operatorname{tr}(XY) = \operatorname{tr}(YX)$ if both products are defined. Let $X = Q^{-1}A$ and $Y = Q$: $\operatorname{tr}(B) = \operatorname{tr}((Q^{-1}A)Q) = \operatorname{tr}(Q(Q^{-1}A)) = \operatorname{tr}(A)$. Thus, $\operatorname{tr}(A) = \operatorname{tr}(B)$. ■

8.6. Are the matrices

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix} \text{ and } \begin{pmatrix} 0 & 2 \\ 4 & 2 \end{pmatrix}$$

similar? Justify.

Proof. No. Let $A = \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 4 & 2 \end{pmatrix}$. If $B = Q^{-1}AQ$ then $AQ = QB$. Letting $Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we can compare entries and see that AQ and QB force $a = b = c = d$, so Q is not invertible due to $\det Q = 0$. Hence, the matrices aren't similar. ■